Assignment -2

1. How can you identify the datatype of a given NumPy array? How do you change the data type of an array?

The datatype of a NumPy array can be identified using the **`dtype`** attribute. It displays the type of elements stored in the array.
To change the datatype of an array, use the **`astype()`** method, which creates a new array with the specified datatype.

Example:

```
import numpy as np

arr = np.array([1, 2, 3, 4])

# Check datatype
print(arr.dtype)

# Change datatype to float
new_arr = arr.astype(float)

print(new_arr)
print(new_arr.dtype)
```

Output:

```
int64
[1. 2. 3. 4.]
float64
```

2. Create an array using NumPy. Then convert a numeric array to a categorical (text) array.

Program:

```
import numpy as np

# Create a numeric array
arr = np.array([10, 20, 30, 40])

print("Numeric Array:")
print(arr)

# Convert numeric array to text (string) array
text_arr = arr.astype(str)

print("Categorical (Text) Array:")
print(text_arr)
```

Output:

```
Numeric Array:
[10 20 30 40]

Categorical (Text) Array:
['10' '20' '30' '40']
```

- `dtype` is used to identify the datatype of a NumPy array.
- `astype()` is used to convert the datatype of an array.
- Converting the numeric array to `str` changes each number into a text (string) value, creating a categorical/text array.